

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Code

```
# FIND-S Algorithm Implementation
```

```
import pandas as pd
```

```
# Training Data
```

```
data = [  
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],  
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],  
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],  
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']  
]
```

```
# Column Names
```

```
columns = [  
    'Sky', 'AirTemp', 'Humidity',  
    'Wind', 'Water', 'Forecast', 'EnjoySport'  
]
```

```
# Create DataFrame
```

```
df = pd.DataFrame(data, columns=columns)
```

```
print("Training Dataset:\n")
```

```
print(df)
```

```

# Separate features and target
X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values

# FIND-S Algorithm
hypothesis = None

print("\nLearning Process:\n")

for i in range(len(X)):
    if y[i] == "Yes":
        if hypothesis is None:
            hypothesis = X[i].copy()
        else:
            for j in range(len(hypothesis)):
                if hypothesis[j] != X[i][j]:
                    hypothesis[j] = '?'

    print(f"After Positive Example {i+1}: {hypothesis}")

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)

```

```
[1]: # FIND-S Algorithm Implementation
```

```
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# Training Data
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
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# Column Names
columns = [
    'Sky', 'AirTemp', 'Humidity',
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# Create DataFrame
df = pd.DataFrame(data, columns=columns)

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print(df)

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X = df.iloc[:, :-1].values
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                if hypothesis[j] != X[i][j]:
                    hypothesis[j] = '?'

    print(f"After Positive Example {i+1}: {hypothesis}")

print("\nFinal Most Specific Hypothesis:")
print(hypothesis)
```

Training Dataset:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Learning Process:

After Positive Example 1: ['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

After Positive Example 2: ['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']

After Positive Example 4: ['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final Most Specific Hypothesis:

['Sunny' 'Warm' '?' 'Strong' '?' '?']

